

A sharp dyadic threshold in a characteristic-two random-cone construction

Alec Kriebel

ORCID: [0009-0001-9320-500X](https://orcid.org/0009-0001-9320-500X)

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Abstract

OpenAI’s October 4 construction uses a projective plane of order $q = 128$ and seven extra letters to obtain a finitely presented torsion-free group with a finite two-dimensional classifying complex and a characteristic-two group algebra that is not directly finite. We determine the least dyadic order $q \geq 4$ at which its unchanged squared turn weights have exponential word-mass decay: it is $q = 32$. A three-class reduction and exact rational certificates prove contraction at 32 and exponential growth at 4, 8, 16. Checking the remaining parameter-dependent estimates yields the same construction on a 532-edge rose. This is a threshold for this specified model, rather than an optimum among all group-algebra examples. Classical HNN embedding also gives a two-generator realization. We spell out the inherited scalar idempotent and nonzero absorbed cyclic projective module, which remain valid over every characteristic-two field. The construction is existential; no numerical matching or group-ring multiplication certificate is claimed.

1 Scope and inherited input

For a torsion-free group G and field K , the scalar idempotent conjecture asks whether $e^2 = e$ in $K[G]$ forces $e \in \{0, 1\}$ [2, Problem 1(c)]. The positive-characteristic version includes characteristic two. The pivotal input here is the exact October 4 manuscript [1, Theorem 1.1], at repository commit [adc7f124](#): it constructs a finitely presented torsion-free group with a finite 2-dimensional $K(G, 1)$ and scalar elements

$$ab = 1, \quad ac = 0, \quad c \neq 0 \quad \text{in } \mathbb{F}_2[G]. \quad (1)$$

The breakthrough and construction belong to OpenAI. Its probabilistic and geometric framework also draws on its September 23 zero-divisor manuscript [7]. Our independent source audits found no substantive gap in the arguments used; their precise scope is recorded in the supplement. The earlier family-197 Lean example has torsion and does not formalize this October 4 theorem. No full formal verification or conventional human refereeing is claimed.

Our contribution is the exact dyadic threshold for the *unchanged* seven-extra model in Section 2, together with the parameter check in Section 3. The defect-idempotent and module consequences were already publicly recorded in the author’s triage [6]; they are included for interpretation, not offered as a new discovery. The two-generator specialization in Section 4 uses classical machinery. We make no first-priority claim and no characteristic-zero, odd-characteristic or reduced group C^* -algebra projection claim.

2 The exact squared-word threshold

Let $q = 2^k$ with $k \geq 2$, $v = q^2 + q + 1$ and $p = (q + 1)/v$. The source's signed alphabet T consists of the v points of $\text{PG}(2, q)$ and seven extra letters. Pair each extra with a distinct ordinary letter, and pair the remaining $v - 7$ ordinary letters among themselves; write $\bar{t}$ for the inverse of t . For a reduced turn $u \neq \bar{t}$, retain exactly the source weights

$$w(t, u) = \begin{cases} (q + 1)^{-1}, & \bar{t}, u \text{ both ordinary,} \\ p, & \bar{t}, u \text{ of different kinds,} \\ 1/2, & \bar{t}, u \text{ both extra,} \end{cases} \quad w(t, \bar{t}) = 0.$$

For $W = t_1 \cdots t_h$, set $P(W) = \prod_{i=1}^{h-1} w(t_i, t_{i+1})$. These are turn-frequency weights, not probabilities assigned to words.

Theorem 1. *In this model, for any allowed inverse pairing, $q = 32$ is the least dyadic $q \geq 4$ for which $M = (w(t, u)^2)$ has spectral radius below one. At $q = 32$, for every $h \geq 1$,*

$$\sum_{W \in T^h} P(W)^2 \leq \frac{5376}{5} \left(\frac{987}{1000} \right)^{h-1}.$$

For each $q \in \{4, 8, 16\}$ the same sum is at least $C_q(503/500)^{h-1}$, where $C_q > 0$ is explicit below.

Proof. Partition the actual letters into E (the seven extras), S (their seven ordinary mates), and O (the remaining $v - 7$ ordinary letters). Vectors constant on these classes are invariant under M . In class order (E, S, O) their exact quotient is

$$Q(q) = \begin{pmatrix} 7p^2 & 6/(q+1)^2 & (v-7)/(q+1)^2 \\ 3/2 & 7p^2 & (v-7)p^2 \\ 7p^2 & 7/(q+1)^2 & (v-8)/(q+1)^2 \end{pmatrix}. \quad (2)$$

For a row in E , its inverse excludes one S successor; for a row in S , its inverse excludes one E successor; for a row in O , its inverse excludes one O successor. This proves the quotient formula for every permitted pairing. Q is strictly positive. Its positive Perron eigenvector lifts to a positive eigenvector of M , so $\rho(M) = \rho(Q)$. The following certificates also prove the bounds directly, without numerical eigenvalues.

At $q = 32$, lift $f = (1, 13/5, 1)$. The three ratios $(Qf)_i/f_i$ are

$$\frac{95146189}{96562235}, \quad \frac{579417}{592826}, \quad \frac{857592557}{869060115}, \quad (3)$$

each strictly below $\lambda = 987/1000$. Hence $Mf \leq \lambda f$. As $\mathbf{1} \leq f$ and $\mathbf{1}^\top f = 5376/5$, nonnegativity gives

$$\mathbf{1}^\top M^{h-1} \mathbf{1} \leq \lambda^{h-1} \mathbf{1}^\top f.$$

The left side is exactly the stated sum over words. Thus the source's decay lemma holds, for example with $\delta = -\frac{1}{4} \log \lambda > 0$ for all sufficiently large h .

For $q = 4, 8, 16$, use $x = (193/500, 1, 97/250)$, with $x \leq \mathbf{1}$. The exact ratios $(Qx)_i/x_i$ are

q	E	S	O
4	480733/303975	26959/21000	500011/305550
8	10342603/9256473	5726739/5329000	10523789/9304434
16	35146373/34932807	3571543/3549000	2078707/2065518

Every entry exceeds $\mu = 503/500 > 1$. Therefore $Mx \geq \mu x$ and

$$\mathbf{1}^\top M^{h-1} \mathbf{1} \geq \mu^{h-1} \mathbf{1}^\top x, \quad C_q = 7 \frac{193}{500} + 7 + (v-7) \frac{97}{250} > 0.$$

This proves $\rho(M) > 1$ for 4, 8, 16, completing the least-order claim. $\square$

The lower bound excludes the source's squared-word decay statement at these orders. It does not prove nonexistence of good matchings or idempotents there, or exclude a modified type model or another method. Order 2 is outside the theorem's domain: the unchanged extra-count prescription would require both $3m-1$ and $3(m-1)$ divisible by four, which is impossible. Exact rational verification is supplied as `verification/verify_parameter.py`.

3 Carrying the construction to order 32

Proposition 2. *The October 4 random-cone construction remains valid at $q = 32$ with the same seven-extra types. It gives a finitely presented torsion-free group G generated by at most 532 elements, a finite 2-dimensional $K(G, 1)$, and the scalar triple (1).*

Proof. Here $v = 1057$, $p = 33/1057$ and $|T| = 1064$. For sufficiently large $m \equiv 1 \pmod{4}$, take m vertices in every line class on side A and $m-1$ on side B , with $n = vm$. Each ordinary outgoing set is its 33-point line. Give the distinguished A vertex all seven extras; for each complement of a Fano line assign that four-element extra set at $(33m-1)/4$ vertices of A and $33(m-1)/4$ of B , evenly among line classes. All other extra sets are empty. Fix $x_B \in B$ before choosing the matchings. The line-capacity coefficient is $7 \cdot 33/(4 \cdot 1057) = 33/604 < 1$, so disjoint assignments exist for all sufficiently large m . Each extra is in four complements and each extra pair in two, while ordinary lines have odd intersection size. Thus the source's exact inverse-slot balances and parity identities persist. Degrees are now 33, 37, 40, with minimum 33 and maximum $d_* = 40$.

The decay input is Theorem 1. The remaining estimates in [1, Sections 3–5] are verified as follows. Choose a fixed $c_0 > 0$ satisfying $c_0 \log(2|T|/p) < 1$ and $2c_0 \log d_* < 1$; $c_0 = 1/100$ works. The short-cycle switching proof then has the same error $r_n = O(\log n + n^{2c_0 \log d_*}) = o(n)$ and supplies the same conditioned edge-prescription bounds. It uses only fixed T, p, d_* , these strict inequalities, and exact slot balance, all of which hold here.

For the diameter argument, a k -vertex set without factor-two expansion forces at least $r = \lceil 33k/2 \rceil$ prescribed edges in a $2k$ -vertex set, with $16.5k \leq r \leq 17k$. The source's union bound becomes

$$\Pr(\text{failure at } k) \leq (C_3(k/n)^{29/2})^k.$$

A fixed small $\eta > 0$ can satisfy $17\eta < p/4$ and $C_3\eta^{29/2} < 1/2$. Summing separately over $k \leq \sqrt{n}$ and $\sqrt{n} < k \leq \eta n$ gives $o(1)$; the first sum uses exponent $29/4 > 0$. The ball-packing argument then gives the same dL component diameter bound, where $L = \lfloor c_0 \log n \rfloor$. The supplemental exact certificate even permits $\eta = 10^{-6}$ with an explicit conservative bound for C_3 .

All bounded-path estimates in Section 4 of the source use the fixed alphabet, the positive minimum turn weight, the just-established prescription bound and exponential squared-word decay. Their constants may change but their order of quantifiers does not: choose the unpaired fraction first, then any fixed path-count, length and comparison bounds. The multiplicity-stage first moments do not require independence between stages. In the planar reduction, deleting at most two incident edges leaves degree at least $31 \geq 3$, so the short-closure lemma is unchanged; the Euler characteristic and separator estimates contain no further projective-plane order requirement. Consequently the probability of a reduced spherical arrangement tends to zero in this conditioned model.

These statements follow by applying the cited proofs with the displayed replacements, rather than treating their originally fixed-128 theorems as unproved general-parameter theorems.

Select one sufficiently large matching outcome with these properties. The deterministic cone criterion in [1, Section 6] uses no q parameter: it gives asphericity and protects the two roots. The parity proposition in its Section 2 yields (1). Coning all graph components onto the 532-edge rose yields a finite 2-dimensional classifying complex. It is homotopy equivalent to the presentation complex on the rose edges with one relator for each non-tree graph edge, so G is generated by at most 532 elements. Torsion-freeness follows as in the source from its finite-dimensional classifying complex. $\square$

For completeness, after choosing an outcome, choose a root and spanning tree in every component, taking x_A and x_B as the roots of their respective components A' , B' , and let w_x be the tree-path word to x . For each oriented non-tree edge $x \xrightarrow{t} y$ impose $w_x t w_y^{-1}$. At the designated-root components A', B' , the witnesses are $a = \sum_{x \in A'} w_x$, $b = \sum_{x \in A'} w_x^{-1}$ and $c = \sum_{y \in B'} w_y^{-1}$. Product-graph parity gives $ab = 1, ac = 0$; root protection makes the identity coefficient of c equal to one. The order-32 incidence plane and inverse pairing have a reproducible finite certificate in the supplement. No numerical m satisfying all asymptotic requirements, matching outcome, relator list, reduced support of e , or target group-ring multiplication certificate is supplied.

4 Classical two-generator realization

The number 532 describes this random-cone recipe, not the smallest possible group generating set. We record explicitly why classical HNN embedding [8, 9] already yields two generators while preserving the stronger finite classifying-space condition.

Proposition 3. *Every group G with a finite 2-dimensional $K(G, 1)$ embeds in a group H generated by two elements with a finite 2-dimensional $K(H, 1)$. If G has an aspherical presentation on $g_1, \dots, g_s$ with r relators, H has an aspherical presentation on two generators with r relators. Torsion-freeness is preserved.*

Proof. Set $B = G * F(x, y)$, and in B define

$$u_0 = y, \quad v_0 = x, \quad u_i = x^{-i} y x^i, \quad v_i = y^{-i} x g_i y^i \quad (1 \leq i \leq s).$$

The u_i freely generate: they are a finite subset of the standard free basis of the kernel of $F(x, y) \rightarrow \mathbb{Z}$, $x \mapsto 1, y \mapsto 0$. Projection killing G sends the v_i to the corresponding free-basis subset with x, y interchanged, and proves they freely generate too. The HNN extension with relations $t^{-1} u_i t = v_i$ therefore embeds B . Its zeroth relation gives $y = x t x^{-1}$; the others give $g_i = x^{-1} y^i t^{-1} x^{-i} y x^i t y^{-i}$. Substituting for y expresses g_i by the word

$$W_i = x^{-1} t x^i t^{-2} x^{-i} t x t^{-1} x^i t^2 x^{-i} t^{-1}.$$

Thus $H = \langle x, t \rangle$, and its presentation is $\langle x, t \mid R_j(W_1, \dots, W_s), 1 \leq j \leq r \rangle$.

Let X be the finite classifying complex for G , with its maximal 1-skeleton tree collapsed. The vertex space $X \vee R_2$ is a finite $K(B, 1)$. Attach the mapping cylinder of a rose with $s + 1$ petals via the two injective maps representing u_i and v_i . This gives a finite complex of dimension at most two for H . Its universal cover is a tree of contractible vertex and edge spaces; successively adjoining finite subtrees gives contractible homotopy pushouts. Sphere maps have compact image contained in a finite subcomplex, and hence a finite subtree, so the cover is contractible. This

proves asphericity. Finite-order elements of an HNN extension are conjugate into its base; the base free product is torsion-free when G is, proving the assertion.

To see that the final two-generator presentation itself is aspherical, first consider the subcomplex having only the $s+1$ HNN defining cells. Each g_i occurs once in its own such cell and in no other such cell, so those generator-cell pairs can be collapsed there; then collapse y with the zeroth defining cell. The resulting retraction to the (x, t) rose sends g_i to W_i . Homotopy invariance of attaching cells carries the original r relators to $R_j(W)$, proving homotopy equivalence with the displayed presentation complex. This uses these specific eliminations, not arbitrary Tietze moves. $\square$

An embedding $G \hookrightarrow H$ induces an injective map $K[G] \rightarrow K[H]$ because distinct group elements remain distinct basis elements. The scalar triple is preserved. No support is shortened or computed by this transport. A torsion-free group generated by at most one element is trivial or infinite cyclic; its group algebra over a field is a domain. Consequently two is the least possible number of generators for such an idempotent example, but the embedding mechanism is classical.

5 Ring and right-module proof

The reduction to a scalar idempotent is classical [3, Proposition `kaplansky_relations`]; the splitting is the regular right-module interpretation of direct finiteness [4, p. 9]. We include the maps to fix handedness.

Proposition 4. *Let R be a nonzero unital ring and suppose (1) holds in R . Then $e = 1 - ba$ is an idempotent other than 0 and 1, and $P = eR$ has all the right-module and K_0 properties in the consequence theorem below. No characteristic assumption is needed for this proposition.*

Proof. Put $f = ba$. Then $f^2 = b(ab)a = f$, so $e^2 = (1 - f)^2 = 1 - 2f + f^2 = 1 - f = e$. Also $ec = c - b(ac) = c \neq 0$, whence $e \neq 0$. If $e = 1$, then $ba = 0$, and $a = a(ba) = 0$ contradicts $ab = 1$ in a nonzero ring. Thus $e \neq 1$.

The complementary idempotents f and e satisfy $fe = ef = 0$ and $f + e = 1$. Therefore $R = fR + eR$. If $x = fr = es$, then $x = fx = fes = 0$, giving the direct sum

$$R_R = baR \oplus eR. \quad (4)$$

Since $baR \subseteq bR$ and $b = bab$, the reverse inclusion holds as well: $baR = bR$. Left multiplication by b is a right R -linear isomorphism $R_R \rightarrow bR$, with inverse left multiplication by a . Indeed $abr = r$ for every r , and $ba(br) = br$ for every $br \in bR$.

Consequently (4) identifies R_R with $R_R \oplus P$. For a fully explicit pair of inverse maps, use

$$\Phi(r) = (ar, er), \quad \Psi(u, p) = bu + p \quad (r, u \in R, p \in eR).$$

The identities $ae = eb = 0$, $ep = p$, and $ab = 1$ give $\Psi\Phi(r) = bar + er = r$ and $\Phi\Psi(u, p) = (u, p)$. They also show $\ker(r \mapsto ar) = eR$; in particular $c \in P$.

The module P is a direct summand of the free right module R_R , hence projective. It is cyclic, generated by e , and nonzero because it contains $e = e \cdot 1 \neq 0$. By definition, $K_0(R)$ is the group completion of the direct-sum monoid of finitely generated projective right modules [5, Chapter II, Section 2]. Hence the isomorphism implies $[R] = [R] + [P]$ and therefore $[P] = 0$ in that abelian group. It does not imply $P = 0$. Rather, $R \oplus 0 \cong R \oplus P$ with $0 \not\cong P$ exhibits failure of projective cancellation. $\square$

Theorem 5. *There is a two-generator finitely presented torsion-free group H with a finite 2-dimensional $K(H, 1)$ and a scalar $e \in \mathbb{F}_2[H]$, $e^2 = e$, $e \neq 0, 1$. For every characteristic-two field K , its same image is nontrivial in $S = K[H]$ and $P = eS$ is a nonzero cyclic projective right module satisfying $S_S \cong S_S \oplus P$ and $[P] = 0$ in $K_0(S)$.*

Proof. Apply Propositions 2 and 3 and transport a, b, c to $\mathbb{F}_2[H]$. Proposition 4 gives $e = 1 - ba$ and the right-module properties. The unique unital embedding $\mathbb{F}_2 \hookrightarrow K$ induces an injective ring map $\mathbb{F}_2[H] \rightarrow K[H]$: the elements of H are a basis over each field. Thus e and $e - 1$ remain nonzero, and all scalar identities persist. Apply the ring proposition again over $K[H]$. $\square$

For $R = \mathbb{F}_2[H]$, vanishing of this one class does not assert $K_0(R) = 0$. The scalar inclusion $\mathbb{F}_2 \rightarrow R$ and augmentation $R \rightarrow \mathbb{F}_2$ split off $K_0(\mathbb{F}_2) = \mathbb{Z}$ from $K_0(R)$. The characteristic-two field embedding supplies no coefficient map to a field of odd characteristic or characteristic zero.

6 Priority, validation and limitations

The published triage [6], commit [f27318d83bd7000ef817957a9a4b3087de28d198](#), already states $e = 1 - ba$, $P = eR$, $R \cong R \oplus P$ and $[P] = 0$, conditional on checking the October 4 input. Those core consequences are duplicated. The present note’s additional result is Theorem 1 and the source adaptation in Proposition 2. The inspected pinned manuscript fixes 128; the relevant companions and triage do not supply this exact calibration or its lower-order obstruction. A finite literature search cannot establish universal priority, and we assert neither a first solution of a named conjecture nor a new embedding or idempotent mechanism.

The supplement retains separate audits of the actual probabilistic and cone-picture proofs, a parameter-specific adversarial audit, primary source identities, exact rational checks, the finite incidence model, and the embedding word calculation. They certify their stated portions; finite computations do not certify an unlisted matching or replace the proof of group existence. No characteristic-zero or odd-characteristic conclusion, matrix-only substitute, nonzero K_0 class or full Lean formalization is inferred.

Disclosure. AI tools were used extensively in research, drafting and verification, including independent automated adversarial audits. This preprint has not undergone conventional human peer review or refereeing. OpenAI is the supplied author of the upstream AI-produced construction; no affiliation or coauthor is inferred.

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